

4. OPTIMIZATION PROBLEMS

BASIC CONCEPTS

■ Definition

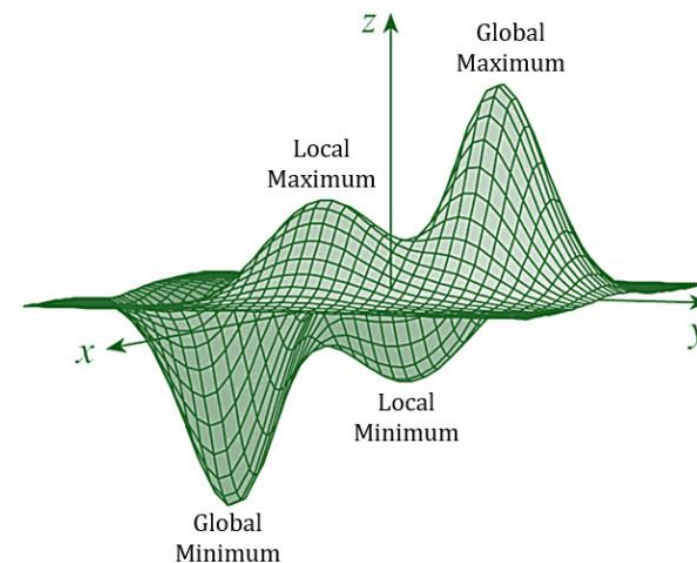
Consider a function $f:D\subset\mathbb{R}^n\rightarrow\mathbb{R}$.

f has a **global maximum** at a point $a\in D$ if: $f(x) \leq f(a) \quad \forall x\in D$ [1]

f has a **global minimum** at a point $a\in D$ if: $f(a) \leq f(x) \quad \forall x\in D$ [2]

f has a **local maximum** if the inequality [1] is verified in an open ball centred in a and contained in D .

f has a **local minimum** if the inequality [2] is verified in an open ball centred in a and contained in D .



- **Formulation of the problem**

In general terms, we can formulate an optimization problem as:

$$\max_{x_1, \dots, x_n} f(x_1, \dots, x_n) \text{ subject to } (x_1, \dots, x_n) \in X$$

where,

$x_1, \dots, x_n$ are the **decision variables**

$X \subset \mathbb{R}^n$ is the **constraint set** (possible values for the variables; it's defined by the problem constraints)

$f(x_1, \dots, x_n)$ is the **objective function** (mathematical description of the objectives of the problem)

BASIC CONCEPTS

We are going to divide optimization problems in three major groups:

➤ **Unconstrained optimization**, where there are no constraints for the problem decision variables, that is $X = \mathbb{R}^n$.

➤ **Optimization with equality constraints**, where the constraint set is defined by independent equality conditions as

$$g_j(x_1, \dots, x_n) = b_j, j = 1, \dots, m \ (m < n)$$

➤ **Optimization with inequality constraints**, where the constraint set is defined by two types of inequality conditions that, in general, can be written as:

$$g_j(x_1, \dots, x_n) \leq b_j, j = 1, \dots, m \ (m < n) \quad (\text{inequality conditions})$$

$$x_i \geq 0, i = 1, \dots, n \quad (\text{non-negativity conditions})$$

4.1 UNCONSTRAINED OPTIMIZATION

OPTIMIZATION IN $\mathbb{R}$ (REVISION)■ **Definition**

Let $f:D \subset \mathbb{R} \rightarrow \mathbb{R}$ and $a \in \text{int}(D)$. If f is differentiable at a and $f'(a) = 0$, then a is called a **critical point** or a **stationary point**.

■ **Theorem**

Let f be a function of class C^n in an interval I and $a \in I$. If $f'(a) = f''(a) = \dots = f^{(n-1)}(a) = 0$ and $f^{(n)}(a) \neq 0$, then

- If n is even and,
 - $f^{(n)}(a) < 0$, f has a local maximum at a
 - $f^{(n)}(a) > 0$, f has a local minimum at a
- If n is odd, $(a, f(a))$ is an inflection point.

UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}$ (REVISION)

■ Examples

Find the local extreme values of the following functions:

(1) $f(x) = \frac{x^3}{3} - 3x^2 + 8x + 6$

$$f'(x) = 0 \Leftrightarrow x^2 - 6x + 8 = 0 \Leftrightarrow x = \frac{6 \pm \sqrt{36 - 32}}{2} \Leftrightarrow x = \frac{6 \pm 2}{2} \Leftrightarrow \boxed{x = 4} \vee \boxed{x = 2} \rightarrow \text{Critical Points}$$

(Candidates for local extreme points)

$$f''(x) = 2x - 6$$

← even order

$$f''(4) = 8 - 6 = 2 > 0. \text{ Therefore, } f(4) \text{ is a } \underline{\text{local minimum}}$$



← even order

$$f''(2) = 4 - 6 = -2 < 0. \text{ Therefore, } f(2) \text{ is a } \underline{\text{local maximum}}$$



UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}$ (REVISION)

■ Examples (cont.)

(2) $f(x) = 2x^3$

$$f'(x) = 0 \Leftrightarrow 6x^2 = 0 \Leftrightarrow \boxed{x=0} \text{ critical point}$$

$$f''(x) = 12x \Rightarrow f''(0) = 0$$

$$f'''(x) = 12 \Rightarrow f'''(0) = 12 > 0 \text{ (not null)} \Rightarrow x=0 \text{ is an } \underline{\text{inflection point}}$$

↑
The fact that is > 0
indicates that f
is increasing at
 $x=0$

f does not have a local extremum
at $x=0$.

UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}$ (REVISION)

■ Examples (cont.)

(3) $f(x) = x^4$

$$f'(x) = 0 \Rightarrow 4x^3 = 0 \Rightarrow x = 0 \quad \leftarrow \text{critical point}$$

$$f''(x) = 12x^2 \Rightarrow f''(0) = 0$$

$$f'''(x) = 24x \Rightarrow f'''(0) = 0$$

$\leftarrow$ even order

$$f^{(4)}(x) = 24 \Rightarrow f^{(4)}(0) = 24 > 0, \text{ Therefore, } f(0) = 0 \text{ is a } \underline{\text{local minimum}}$$

$\cup$

UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}^n$

Let's begin by considering $n=2$, that is a function with two variables $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ that has a local maximum at a certain point (a, b) .

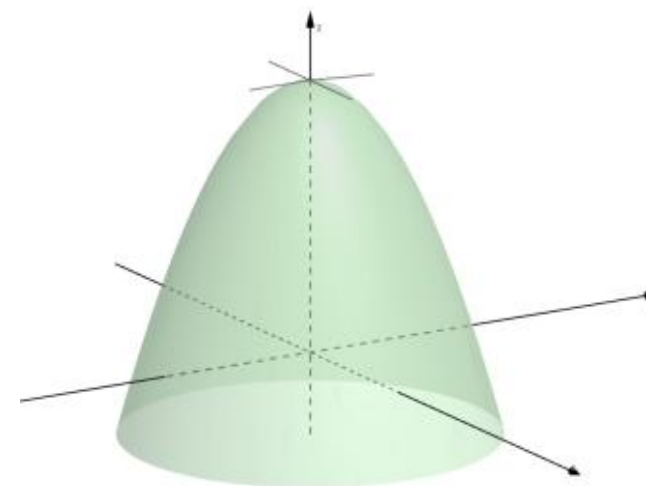
If there exists a local maximum of f at (a, b) , then an infinitesimal deviation from the point (a, b) in any possible direction will result in an infinitesimal variation in the function.

In mathematical terms, it means that

$$df(a, b) = 0, \quad \forall dx, dy$$

$$\frac{\partial f}{\partial x}(a, b)dx + \frac{\partial f}{\partial y}(a, b)dy = 0, \quad \forall dx, dy$$

$$\frac{\partial f}{\partial x}(a, b) = 0 \quad \text{and} \quad \frac{\partial f}{\partial y}(a, b) = 0$$



UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}^n$ ■ **1st Order Conditions**

Let $f: D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function in $\text{int}(D)$ and let $a \in \text{int}(D)$. If f has a local extremum at a , then $\nabla f(a) = \mathbf{0}$.

In these conditions, we say that a is a **critical point** or a **stationary point**.

■ **2nd Order Conditions**

Let $f: D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a function of class C^2 in an open ball centered in $a \in \text{int}(D)$ and let a be a critical point of f .

I. If $d^2f(a) < 0$ then $f(a)$ is a local maximum of f .

II. If $d^2f(a) > 0$ then $f(a)$ is a local minimum of f .

III. If $d^2f(a) > 0$ and $d^2f(a) < 0$, then $f(a)$ is not an extremum. $(a, f(a))$ is a **saddle point**.

UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}^n$ ■ **2nd Order Conditions (cont.)**

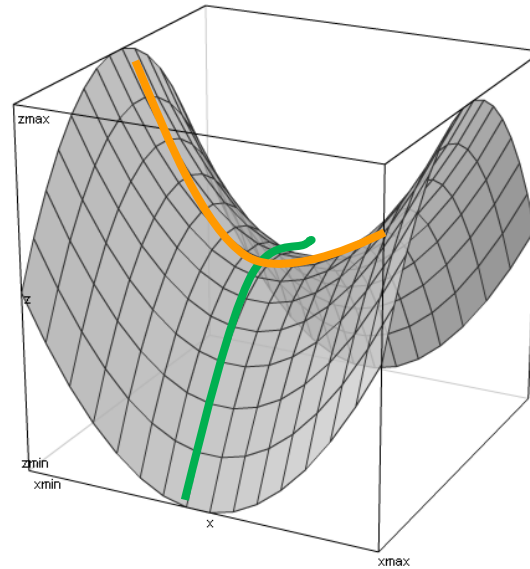
- IV. If $D^2f(a) \leq 0$ [or ≥ 0], that is, if $D^2f(a)$ is a negative [or positive] semidefinite quadratic form, then we have to analyze the singular directions (directions that null $D^2f(a)$):
- if there is any singular direction in which the first directional derivative that doesn't null is of odd order or of even order but of opposite sign of $D^2f(a)$ outside the singular directions, then $f(a)$ is not an extreme.
 - If the directional derivatives continue to be null or if the first one that does not null is of even order and of the same sign of $D^2f(a)$ outside the singular directions, then nothing can be concluded.
- V. If $D^2f(a) = 0$, we need to analyze the higher-order differentials:
- if the first differential that is not null is of odd order, then $f(a)$ is not an extreme.
 - If the first differential that is not null is of even order, then we consider the conditions given in I, II and III,

■ **Remark**

The cases (IV) and (V) of the 2nd order conditions will not be studied!

UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}^n$

The following graph illustrates a case of a critical point that is not a local extremum, it is a saddle point.



UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}^n$

■ Remark

$$d^2f(a,b) = \frac{\partial^2 f}{\partial x^2}(a,b) dx^2 + 2 \frac{\partial^2 f}{\partial x \partial y}(a,b) dx dy + \frac{\partial^2 f}{\partial y^2}(a,b) dy^2 = [dx \ dy] \overset{H(a,b)}{\begin{bmatrix} \frac{\partial^2 f}{\partial x^2}(a,b) & \frac{\partial^2 f}{\partial x \partial y}(a,b) \\ \frac{\partial^2 f}{\partial y \partial x}(a,b) & \frac{\partial^2 f}{\partial y^2}(a,b) \end{bmatrix}} \begin{bmatrix} dx \\ dy \end{bmatrix}$$

Remember that the second-order differential, $d^2f(a_1, \dots, a_n)$, is a quadratic form in dx_1 and dx_n that has the Hessian matrix at $(a_1, \dots, a_n)$ as associated symmetric matrix.

$$d^2f(a_1, \dots, a_n) = [dx_1 \quad \dots \quad dx_n] H(a_1, \dots, a_n) \begin{bmatrix} dx_1 \\ \vdots \\ dx_n \end{bmatrix}$$

So, to study the sign of $d^2f(a_1, \dots, a_n)$ is to classify it as quadratic form - positive definite or negative definite or indefinite (semidefinite cases will not be studied).

QUADRATIC FORMS (REVISION)

■ Definition

Quadratic forms are polynomials that only have terms of degree 2. For example:

<u>Polynomial notation</u>	$q(x, y) = x^2 - 2xy + 2y^2$
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<u>Matrix notation</u>	$q(x, y) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$
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<u>Polynomial notation</u>	$q(x, y, z) = -2x^2 + 2xz - y^2 - 3z^2$
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<u>Matrix notation</u>	$q(x, y, z) = \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} -2 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
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QUADRATIC FORMS (REVISION)

■ Definition

Consider the matrix $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$. The **principal minors** of A are:

$$\text{➤ } |A_1| = a_{11}$$

$$\text{➤ } |A_2| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$\text{➤ } |A_3| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

...

$$\text{➤ } |A_n| = |A|$$

QUADRATIC FORMS (REVISION)

- **Classification of a quadratic form using the principal minors of the associated symmetric matrix**

Let q_A a quadratic form with associated symmetric matrix A . Then

- If all principal minors of A are positive, then q_A is a **positive definite quadratic form**.
- If the principal minors of A alternate in sign, being the first one negative ($|A_1| < 0, |A_2| > 0, |A_3| < 0, \dots$) then q_A is a **negative definite quadratic form**.
- If all minors of A are greater or equal to zero and at least one of them is zero, then q_A is a positive semidefinite quadratic form.
- If all minors of A of even order are greater or equal to zero and all minors of A of odd order are less or equal to zero, and at least one of them is zero, then q_A is a negative semidefinite quadratic form.
- If none of the other conditions are verified, then q_A is an **indefinite quadratic form**.

UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}^n$

■ Examples

(1) $f(x, y) = 8x^3 + 2xy - 3x^2 + y^2 + 1$

First-order conditions: $\nabla f(x, y) = (0, 0) \Leftrightarrow \begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 24x^2 + 2y - 6x = 0 \\ 2x + 2y = 0 \end{cases} \Leftrightarrow \begin{cases} 24x^2 - 8x = 0 \\ y = -x \end{cases} \Leftrightarrow \begin{cases} 8x(3x-1) = 0 \\ - \end{cases}$

$$\Leftrightarrow \begin{cases} x=0 \\ y=0 \end{cases} \vee \begin{cases} x=\frac{1}{3} \\ y=-\frac{1}{3} \end{cases}$$

$(0, 0)$ and $(\frac{1}{3}, -\frac{1}{3})$ are critical points (candidates for local extreme points)

Second-order conditions: We must study the sign of the second order differential at the critical points.

$$H(x, y) = \begin{bmatrix} 48x-6 & 2 \\ 2 & 2 \end{bmatrix}$$

$\rightarrow H(\frac{1}{3}, -\frac{1}{3}) = \begin{bmatrix} 10 & 2 \\ 2 & 2 \end{bmatrix}$ Principal minors:
 $|H_1| = 10 > 0$

$$|H_2| = \begin{vmatrix} 10 & 2 \\ 2 & 2 \end{vmatrix} = 20 - 4 = 16 > 0$$

$d^2f(\frac{1}{3}, -\frac{1}{3})$ is a positive definite quadratic form, that is $d^2f(\frac{1}{3}, -\frac{1}{3}) > 0$. So, $f(\frac{1}{3}, -\frac{1}{3})$ is a local minimum.

$\rightarrow H(0, 0) = \begin{bmatrix} -6 & 2 \\ 2 & 2 \end{bmatrix}$ Principal minors:
 $|H_1| = -6 < 0$
 $|H_2| = \begin{vmatrix} -6 & 2 \\ 2 & 2 \end{vmatrix} = -12 - 4 = -16 < 0$

$d^2f(0, 0)$ is an indefinite quadratic form, so, $(0, 0)$ is a saddle point.

UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}^n$

■ Examples

(2) $f(x, y) = x + 2ey - e^x - e^{2y}$

First-Order Conditions: $\nabla f(x, y) = (0, 0) \Leftrightarrow \begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 1 - e^x = 0 \\ 2e - 2e^{2y} = 0 \end{cases} \Leftrightarrow \begin{cases} e^x = 1 \Leftrightarrow \boxed{x=0} \\ e^{2y} = e \Leftrightarrow 2y=1 \Leftrightarrow \boxed{y=\frac{1}{2}} \end{cases}$

$(0, \frac{1}{2})$ is a critical point of f

Second-order Conditions: we must study the sign of $d^2f(0, \frac{1}{2})$

$$H(x, y) = \begin{bmatrix} -e^x & 0 \\ 0 & -4e^{2y} \end{bmatrix}$$

$$H(0, \frac{1}{2}) = \begin{bmatrix} -1 & 0 \\ 0 & -4e \end{bmatrix}$$

Principal Minors:

$$|H_1| = -1 < 0$$

$$|H_2| = \begin{vmatrix} -1 & 0 \\ 0 & -4e \end{vmatrix} = 4e > 0$$

As the principal minors of $H(0, \frac{1}{2})$ alternate in sign starting with negative, we conclude that $d^2f(0, \frac{1}{2})$ is a negative definite quadratic form, that is $d^2f(0, \frac{1}{2}) < 0$.

So, $f(0, \frac{1}{2}) = e - 1 - e = \boxed{-1}$ is a local maximum of f

UNCONSTRAINED OPTIMIZATION IN $\mathbb{R}^n$

■ Examples

(3) $f(x, y, z) = 2x^2 + xy + 4y^2 + xz + z^2 + 2$

First-order Conditions:

$$\nabla f(x, y, z) = (0, 0, 0) \Leftrightarrow \begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \\ \frac{\partial f}{\partial z} = 0 \end{cases} \Leftrightarrow \begin{cases} 4x + y + z = 0 \\ x + 8y = 0 \\ x + 2z = 0 \end{cases} \Leftrightarrow \begin{cases} - \\ x = -8y \\ -8y + 2z = 0 \end{cases} \Leftrightarrow \begin{cases} -32y + y + 4y = 0 \\ x = -8y \\ z = 4y \end{cases} \Leftrightarrow \begin{cases} -27y = 0 \Rightarrow y = 0 \\ x = 0 \\ z = 0 \end{cases}$$

Second-order Conditions: study the sign of $d^2f(0,0,0)$

$$H(x, y, z) = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 8 & 0 \\ 1 & 0 & 2 \end{bmatrix} \quad (= H(0,0,0))$$

Principal minors of $H(0,0,0)$:

$$|H_1| = 4 > 0$$

$$|H_2| = \begin{vmatrix} 4 & 1 \\ 1 & 8 \end{vmatrix} = 32 - 1 = 31 > 0$$

$$|H_3| = \begin{vmatrix} 4 & 1 & 1 \\ 1 & 8 & 0 \\ 1 & 0 & 2 \end{vmatrix} = 64 + 0 + 0 - 8 - 0 - 2 = 54 > 0$$

So, $d^2f(0,0,0)$ is a positive definite quadratic form, that is $d^2f(0,0,0) > 0$.

Therefore, $f(0,0,0) = 2$ is a local minimum